

# Digital Systems Project Assignment 1

## Digital Watch: Settable Time

<https://chipverify.com/verilog/verilog-data-types>

### What values do variables hold ?

Almost all data-types can only have one of the four different values as given below except for `real` and `event` data types.

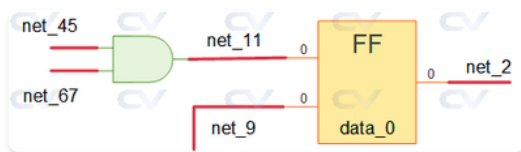
|   |                                                        |
|---|--------------------------------------------------------|
| 0 | represents a logic zero, or a false condition          |
| 1 | represents a logic one, or a true condition            |
| x | represents an unknown logic value (can be zero or one) |
| z | represents a high-impedance state                      |

### Nets and Variables

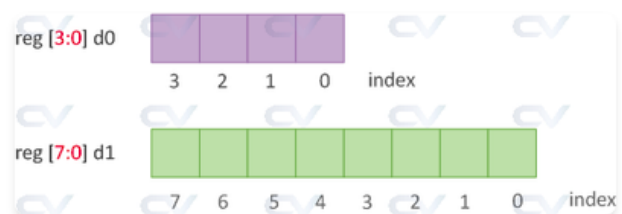
*Nets* and *variables* are the two main groups of data types which represent different hardware structures and differ in the way they are assigned and retain values.

#### Nets

*Nets* are used to connect between hardware entities like logic gates and hence do not store any value on its own. In the image shown below, a net called `net_11` is used to connect between the output of the AND gate to the first input of the flip-flop called `data_0`. In a similar way, the two inputs of the AND gate are connected to nets `net_45` and `net_67`.



#### Variables



A variable on the other hand is an abstraction of a data storage element and can hold values. A flip-flop is a good example of a storage element.

<https://chipverify.com/systemverilog/systemverilog-data-type-logic>

### logic: A 4-State Data Type

`logic` is a 4-state data type — its value set (0, 1, X, Z) is identical to `reg`'s. It isn't, by itself, a variable or a net; it's just the type. When you write a plain declaration like `logic a;` with no other keyword, SystemVerilog implicitly treats that as `var logic a;` — a **variable** of type `logic`.

## 5. Development Cycle

The golden rule of software development — which applies equally to Verilog development — is to write a *small* amount of code, compile and test it thoroughly, finalise its documentation, and then leave it alone unless you need to fix a bug down the track. We use this approach to build small, reusable modules. Near the end, when these modules must be assembled, we shift to a feature-layered strategy: add a small feature, compile and test thoroughly, then move on to the next feature.

Thousands of lines of automated test code have been written for you. In real life, you or your team would be responsible for developing and maintaining this test code. As the saying goes, “*if it is not tested, it is broken.*” Test code is critical: it improves reliability, prevents costly defects, makes changes safer, and serves as living documentation.

Each write-compile-test loop follows what we call the *development cycle*. This cycle is exemplified below.

**Conventions** A good project has an agreed set of conventions.

Each module should live in its own file within the `rtl/` subdirectory, with the filename matching the module name. Names should be descriptive and should only be abbreviated when the full name would be overly long. Well-established abbreviations such as `clk` and `rst` are acceptable, provided they are used consistently.

Naming conventions are as follows. (Commonly, `localparam` constants are also capitalised, but our particular linter disagrees.)

| Kind                 | Convention | Example                 |
|----------------------|------------|-------------------------|
| Modules and signals  | snake_case | next_state, tick_enable |
| parameter constants  | ALL_CAPS   | MAX, WIDTH              |
| localparam constants | CamelCase  | TicksPerSecond          |

### 5.1. Module Specification

For each module, you will be given its declaration together with a description of the required behaviour. Follow the specification exactly, or the automated tests will fail.

### 5.2. Write the Module

Whenever you are asked to write a new module — starting now, for the `seven_segment` specification above — do the following:

### 5.3. Lint, Compile then Generate Waveforms

In this assignment we use the terminal for four things (in the next assignment, it will also be used to run a formal verification tool):

1. More thorough linting (`verilator -Wall --lint-only`).
2. Compiling code (`iverilog -Wall -g2012`).
3. Running test benches (`iverilog -Wall -g2012`, then `./a.out`).
4. Running the automated test suite (`pytest`).

**Lint** Run: `verilator -Wall --lint-only rtl/seven_segment.sv`

It reports 20 warnings before stopping. These are warnings rather than errors: the code is valid SystemVerilog and will compile. **Do not mistake “warning” for “harmless”: a warning usually means the tool has noticed you doing something you did not intend.** Most of the warnings below correspond to genuine bugs. Even the one that does not, noted as it arises, should still be eliminated: being explicit about your intent is good practice, and the goal is always zero warnings.

## Report 1

Explain why Verilator accepts `if` (blank) but not `if` (ACTIVE\_LOW).

```
%Warning-WIDTHTRUNC: rtl/seven_segment.sv:42:32: Logical operator COND expects 1 bit on
the Conditional Test, but Conditional Test's VARREF 'ACTIVE_LOW' generates 32 bits.
```

```
42 |     assign segments = ACTIVE_LOW ? !segments_active_high : segments_active_high;
    |                               ^
```

The if-conditional statement requires its expression to resolve to either True (1) or False (0). In other words, a binary outcome from 1-bit. It does not encode three or more outcomes by itself.

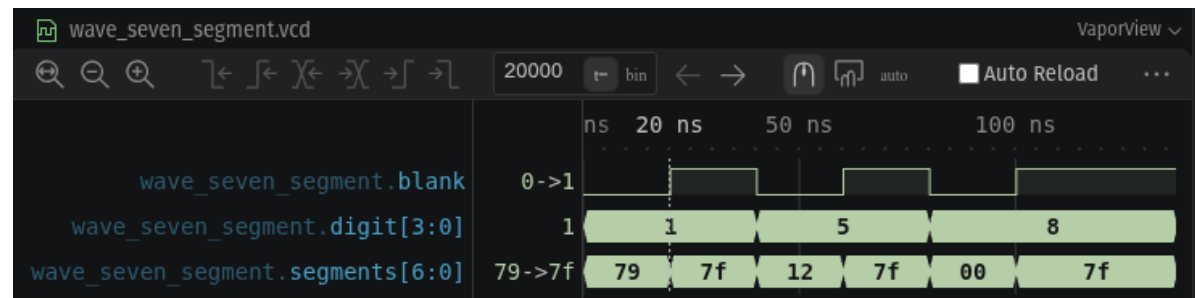
ACTIVE\_LOW is declared as an `int`, which is a 2-state, 32-bit signed integer, and it can contain values other than 0 or 1. Although any non-zero values gets treated as True, the if condition still expects only single-bits and not 32-bits for the conditional tests

```
rtl > seven_segment.sv > SystemVerilog - Language Support > seven_segment
1 // Seven-segment display decoder for hexadecimal digits (0-F).
2 //
3 // Parameters:
4 //   ACTIVE_LOW:          - 1 for active-low LEDs, 0 for active-high.
5 //
6 // Ports:
7 //   digit      [3:0]    - Hexadecimal digit to display (0x0 to 0xF).
8 //   blank      [0]      - When high, all segments are turned off.
9 //   segments   [6:0]    - Segment outputs [g,f,e,d,c,b,a].
10 //
11 // logic datatype default length is 1-bit long
12
13 `timescale 1ns / 1ps
14
15 module seven_segment #(
16     parameter int ACTIVE_LOW = 1
17 ) (
18     input logic [3:0] digit,
19     input logic blank,
20     output logic [6:0] segments
21 );
22
23     logic [6:0] segments_active_high;
24
25     always_comb
26     if (blank) segments_active_high = 7'b0000000; // All segments off
27     else
28         unique case (digit)
29             4'h0: segments_active_high = 7'b0111111; // a,b,c,d,e,f
30             4'h1: segments_active_high = 7'b0000110; // b,c
31             4'h2: segments_active_high = 7'b1011011; // a,b,d,e,g
32             4'h3: segments_active_high = 7'b1001111; // a,b,c,d,g
33             4'h4: segments_active_high = 7'b1100110; // b,c,f,g
34             4'h5: segments_active_high = 7'b1101101; // a,c,d,f,g
35             4'h6: segments_active_high = 7'b1111101; // a,c,d,e,f,g
36             4'h7: segments_active_high = 7'b0000111; // a,b,c
37             4'h8: segments_active_high = 7'b1111111; // a,b,c,d,e,f,g
38             4'h9: segments_active_high = 7'b1101111; // a,b,c,d,f,g
39             4'ha: segments_active_high = 7'b1110111; // a,b,c,e,f,g
40             4'hb: segments_active_high = 7'b1111100; // c,d,e,f,g
41             4'hc: segments_active_high = 7'b0111001; // a,d,e,f
42             4'hd: segments_active_high = 7'b1011110; // b,c,d,e,g
43             4'he: segments_active_high = 7'b1111001; // a,d,e,f,g
44             4'hf: segments_active_high = 7'b1110001; // a,e,f,g
45         endcase
46
47     assign segments = ACTIVE_LOW != 0 ? ~segments_active_high : segments_active_high;
48
49 endmodule
50
```

```

root@4034a70d84cc:/workspace# verilator -Wall --lint-only rtl/seven_segment.sv
root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/seven_segment.sv
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/seven_segment.sv tb/wave_seven_segment.v
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
root@4034a70d84cc:/workspace# ./a.out
VCD info: dumpfile wave_seven_segment.vcd opened for output.
tb/wave_seven_segment.v:41: $finish called at 120000 (1ps)
root@4034a70d84cc:/workspace#

```



## 5.4. Run the Automated Tests

From the terminal, run `pytest tests/test_seven_segment.py` to test only this module, or `pytest` to test all modules. Tests for unwritten modules will be skipped.

Check that all tests for `tests/test_seven_segment.py` pass. If they are reported as skipped, the file is either misnamed or not in the `rtl` subdirectory.

If any tests fail, work through the following steps:

- Scroll up through the `pytest` output to identify which test failed.
- Read the comments in `tests/cocotb/tb_seven_segment.py` to understand what the failing test checks.
- Focus on the corresponding part of your module.

If you are still stuck, see the next section.

```

root@4034a70d84cc:/workspace# pytest tests/test_seven_segment.py
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0
rootdir: /workspace
configfile: .pytest.toml
plugins: metadata-3.1.1, html-4.2.0
collected 2 items

tests/test_seven_segment.py .. [100%]

===== 2 passed in 0.57s =====
root@4034a70d84cc:/workspace#

```

## 5.5. Write Your Own Test Benches

If your module is still not working, write your own test bench. Although the error dump from `pytest` is verbose, and much of it may be unintelligible, scroll through it for messages that make sense. There will usually be a comment explaining what the test was checking, followed by a statement of why it failed. Target that same behaviour in your test bench: your aim is to reproduce the fault.

If reproducing the faulty behaviour requires many steps, simplify it to the fewest steps that still exhibit the fault. Focus on one bug at a time.

Once you have a test bench that reproduces the bug, trace your code through the same steps. This should reveal where your code behaves unexpectedly.

## 5.6. Fulfil Report and Oral Preparation Obligations

The following obligations apply to every module, including `seven_segment`.

### Report 2

For every module, include a screenshot of the waveforms and briefly explain why they are correct, pointing to specific features in the traces.

### Additional Oral Preparation 1

For every module, be able to describe its behaviour precisely for all input combinations.

### Additional Oral Preparation 2

Be able to explain the development cycle and the project conventions in detail.

## 5.7. Backup

After each module passes all its tests, back up your code and report.

### Report 3

At the start of your report, describe your backup strategy.

- Fork Repo
- Push commits

## 6. Time Display

By the end of this section, the seven-segment displays will show hours, minutes, and seconds, incrementing once per second like a clock. Switches select faster rates. Time is in 24-hour format.

### 6.1. State Encoding

Before writing any code, we must decide how to represent the current time in hardware. There are at least four reasonable choices.

1. **One counter (clock cycles).** Count the total number of rising edges since midnight. With a 50 MHz clock, a rising edge occurs every 20 ns.
2. **One counter (seconds).** Count the total number of seconds elapsed since midnight.
3. **Three counters.** Maintain separate counters for hours (0–23), minutes (0–59), and seconds (0–59).
4. **Six counters.** Maintain a separate counter for each display digit: the tens and units of hours, minutes, and seconds.

The choice of state encoding directly influences the number of flip-flops required, the complexity of the next-state logic, and the complexity of the output logic.

The encoding choice also influences the time-setting hardware developed later. Time-setting works by cycling through each unit (seconds, minutes, and hours) and letting the user increment or decrement each independently.

This project adopts the three-counter encoding: separate counters for hours, minutes, and seconds.

#### Additional Oral Preparation 3

Be able to explain all four state encoding options and the trade-offs between them, and justify why the three-counter encoding was chosen.

$$1 \text{ Day} = 24 \text{ Hours} = 1440 \text{ Minutes} = 86400 \text{ Seconds}$$

Option 1: One Counter (Clock Cycles)

CLOCK\_50 is 50MHz clock, this means if the Counter is counting total number of rising edges since midnight, there are  $(50 \times 10^6) \times (86400)$  rising edges =  $4.32 \times 10^{12}$  rising edges.

Therefore, the Counter needs  $\log_2(4.32 \times 10^{12}) = 41.97 = 42$ -bit length Rising-Edge Counter

Total FlipFlops required = 42

Option 2: One Counter (Seconds)

Counter requires a Modulo-50MHz counter to add 1 to itself. Max value is 86400 (seconds).

Seconds Counter needs  $\log_2(86400) = 16.40 = 17$ -bit length Seconds Counter

Total FF required = 17

Requires continuous assignment logic to factor states of Seconds Counter into required HH:MM:SS.

Option 3: Three Counters (Hours, Minutes and Seconds)

Separate Counters for Hours (0-23), Minutes (0-59) and Seconds (0-59).

- Seconds Counter needs Modulo-60 to reset to 0 and adds 1 to Minutes Counter.

-- Needs  $\log_2(59) = 5.88 = 6$ -bit length Seconds Counter.

- Minutes Counter similar to Seconds Counter: Modulo-60 to reset to 0 and adds 1 to Hours Counter.

-- Needs  $\log_2(59) = 5.88 = 6$ -bit length Minutes Counter.

- Hours Counter needs Modulo-24 counter and resets to 0 for next day.

-- Needs  $\log_2(23) = 4.52 = 5$ -bit length Hours Counter.

- Total FF required =  $6 + 6 + 5 = 17$

- The 3 Counters encode the HH:MM:SS states directly. No need for additional logic

Option 4: Six Counters (Tens and Ones Counters for Hours, Minutes and Seconds)

Can use parametrised Counters, just need different values for each of the 6 Counters

Max 5/9 for Tens/Ones for Minutes and Seconds. FF =  $4+4+4+4 = 16$

Max 2/9 for Tens/Ones for Hours, with additional logic condition of 2/3. FF =  $2+4 = 6$

Total FF required =  $16 + 6 = 22$

Although only 2 parametrised modules are required (Tens and Minutes) to represent HH:MM:SS, additional logic is required to ensure HH:MM:SS properly loops back to 0, with HH having more.

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3. Running test benches (`iverilog -Wall -g2012`, then `./a.out`).
4. Running the automated test suite (`pytest`).

## 6.2. A Parametrised Up-Down Counter with Enable

**You must follow the development cycle for every module** (Section 5). Among other things, this includes eliminating all Verilator lint warnings and errors, generating and including waveforms in your report, and ensuring that all automated tests pass.

---

```
1  module up_down_counter #(
2      parameter int MAX    = 2,
3      parameter int WIDTH = 2
4  ) (
5      input logic clk,
6      input logic enable,
7      input logic up,
8      output logic [WIDTH-1:0] count
9  );
```

---

The count is initialised to 0.

When enable is low, count does not change.

When enable is high, count increments if up is high and decrements if up is low, wrapping from MAX to 0 on increment and from 0 to MAX on decrement.

**Rationale** To edit time conveniently, we must be able to increment and decrement the count.

**Implementation Notes** It is conventional<sup>1</sup> to gate enable inside the `always_ff` block, even though it could instead be absorbed into the next-state logic.

---

```
1  always_ff @(posedge clk) if (enable) count <= next_count;
```

---

Use `always_comb` for the next-state logic.

Verilator will flag bit-width mismatches; use SystemVerilog's cast notation to adjust the width of a value explicitly.

---

<sup>1</sup>This mirrors real flip-flops, many of which have a dedicated clock-enable pin.

```

rtl > up_down_counter.sv > Verible > up_down_counter
1 // Up_Down_Counter
2 // To act as a standard Counter, whose modulus value is determined by the
3 // MAX parameter argument that was passed into it. The counter will count up
4 // when the up input is high, and count down when the up input is low. The
5 // counter will wrap back to 0 when the maximum value is reached, and wrap
6 // back to the maximum value when the minimum value is reached for up and
7 // down counting, respectively.
8 //
9 // Parameters:
10 // MAX - Maximum count value (default: 2).
11 // WIDTH - Width of the count output (default: 2).
12 //
13 // Ports:
14 // clk - Clock input.
15 // enable - When high, the counter is enabled.
16 // up - When high, the counter counts up; when low, counts down.
17 //
18
19 `timescale 1ns / 1ps
20
21 module up_down_counter #(
22     parameter int MAX = 2,
23     parameter int WIDTH = 2
24 ) (
25     input logic clk,
26     input logic enable,
27     input logic up,
28     output logic [WIDTH-1:0] count = WIDTH'(0)
29 );
30
31 // Declare local parameters and signals
32 // Cast input `MAX` parameter to local `Max` parameter
33 localparam logic [WIDTH-1:0] Max = WIDTH'(MAX);
34 logic [WIDTH-1:0] next_count;
35
36 // State, and Output Logic
37 always_ff @(posedge clk) if (enable) count <= next_count;
38
39 // Next state logic
40 always_comb begin
41     if (up) begin // up=1 -> Count up
42         if (count < Max) next_count = count + WIDTH'(1);
43         else next_count = WIDTH'(0); // Wrap to 0 when Max
44     end else begin // else up=0 -> Count down
45         if (count > WIDTH'(0)) next_count = count - WIDTH'(1);
46         else next_count = Max; // Wrap to Max when 0
47     end
48 end
49

```

up\_down\_count

```

root@4034a70d84cc:/workspace# verilator -Wall --lint-only rtl/up_down_counter.sv
root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/up_down_counter.sv
root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/up_down_counter.sv tb/wave_up_down_counter.v
root@4034a70d84cc:/workspace# ./a.out
VCD info: dumpfile wave_up_down_counter.vcd opened for output.
tb/wave_up_down_counter.v:26: $finish called at 220000 (1ps)
root@4034a70d84cc:/workspace#

```



```

root@4034a70d84cc:/workspace# pytest tests/test_up_down_counter.py
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0
rootdir: /workspace
configfile: .pytest.toml
plugins: metadata-3.1.1, html-4.2.0
collected 3 items

tests/test_up_down_counter.py ... [100%]

===== 3 passed in 0.93s =====
root@4034a70d84cc:/workspace#

```

### 6.3. Binary to Binary-Coded Decimal

```
1 module binary_to_bcd (  
2     input logic [6:0] bin,    // binary input, 0-99  
3     output logic [3:0] tens,  // decimal tens digit (BCD)  
4     output logic [3:0] ones   // decimal ones digit (BCD)  
5 );
```

For any input value `bin` in the range 0 to 99, the output `tens` represents the decimal tens digit of `bin`, and the output `ones` represents the decimal ones digit.

**Rationale** Displaying hours, minutes and seconds on seven-segment displays requires a binary to binary-coded decimal converter.

#### Report 4

Explain what binary-coded decimal (BCD) is. You may need to research it.

**Implementation Notes** The linter expects the same bit width on both sides of a division. Since `bin` is 7 bits wide, we must write `bin / 7'd10`.

Writing `assign tens = bin / 7'd10` will still produce a warning: the LHS (left-hand side) is 4 bits wide, but the RHS (right-hand side) is 7 bits wide. As we genuinely want to discard the leading bits, we must explicitly convert the result to 4 bits.

```
1 assign tens = 4'(bin / 7'd10);
```

This may seem pedantic when you want to finish the assignment quickly, but in mission-critical designs, you must always be conscious of bit width. The linter forces you to be explicit about your intent.

When viewing the waveforms, set the viewer format for `bin` to decimal.

<https://www.electronics-tutorials.ws/binary/binary-coded-decimal.html>

### What is a Binary Coded Decimal Number?

**Binary Coded Decimal (BCD)** is another type of binary encoding used in digital and electronic circuits. Unlike the standard binary numbering system, which converts the entire number, BCD converts each digit individually, making it ideal for 7-segment displays and input keyboards.

### BCD For Driving Output Displays

Nevertheless, the use of a BCD coding system in both microelectronics and computer systems is particularly useful in situations where the binary coded decimal is intended to be displayed on one or more 7-segment LED or LCD displays and there are many popular integrated circuits available that are configured to give a BCD output or outputs.

### Binary Coded Decimal Summary

We have seen here that **Binary Coded Decimal** or **BCD** is simply the 4-bit binary code representation of a decimal number. Each decimal digit replaced in the integer and fractional parts with its binary equivalent. BCD Code uses four bits to represent the 10 decimal digits of 0 to 9.

The main advantage of the **Binary Coded Decimal** system is that it is a fast and efficient system to convert the decimal numbers into binary numbers as compared to the pure binary system. But the BCD code is wasteful as many of the 4-bit states (10-to-16) are not used.

This because with 4 bits, we can represent sixteen numbers (0000 to 1111) in binary. However, in the BCD code only the first ten are used (0000 to 1001). The remaining six code combinations, i.e., 1010 to 1111 (A to F in Hex) are invalid in the binary coded decimal numbering format.

```

rtl > binary_to_bcd.sv > ...
1 // Binary to Binary-Coded Decimal (BCD) Converter
2 // Displaying hours, minutes, and seconds on seven-segment displays
3 // requires a binary to binary-coded decimal (BCD) converter.
4 // This module converts a 7-bit binary input (0-99) into two
5 // 4-bit BCD outputs representing the tens and ones digits.
6 //
7 // Note: Module allows conversion up to d'127 even though the
8 // maximum expected input is d'99. The BCD outputs will be d'12 and
9 // d'7 for an input of d'127.
10 //
11 // Parameters:
12 // None
13 //
14 // Ports:
15 // bin [6:0] - Binary input b'(0-110_0011) for d'(0-99).
16 // tens [3:0] - BCD output for the tens digit.
17 // ones [3:0] - BCD output for the ones digit.
18 //
19
20 `timescale 1ns / 1ps
21
22 module binary_to_bcd (
23     input logic [6:0] bin, // binary input, b'(0-110_0011)
24     output logic [3:0] tens, // decimal tens digit (BCD)
25     output logic [3:0] ones // decimal ones digit (BCD)
26 );
27
28     assign tens = 4'(bin / 7'd10); // Calculate the tens digit
29     assign ones = 4'(bin % 7'd10); // Calculate the ones digit
30
31 endmodule
32

```



```

• root@4034a70d84cc:/workspace# verilator -Wall --lint-only rtl/binary_to_bcd.sv
• root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/binary_to_bcd.sv
• root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/binary_to_bcd.sv tb/wave_binary_to_bcd.v
• root@4034a70d84cc:/workspace# ./a.out
VCD info: dumpfile wave_binary_to_bcd.vcd opened for output.
tb/wave_binary_to_bcd.v:37: $finish called at 100000 (1ps)
❖ root@4034a70d84cc:/workspace#

```



```

• root@4034a70d84cc:/workspace# pytest tests/test_binary_to_bcd.py
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0
rootdir: /workspace
configfile: .pytest.toml
plugins: metadata-3.1.1, html-4.2.0
collected 1 item

tests/test_binary_to_bcd.py . [100%]

===== 1 passed in 0.31s =====
❖ root@4034a70d84cc:/workspace#

```

## 6.4. Cascaded Hour-Minute-Second Counter with Enable

```
1 module hms_counter #(
2     parameter int N_HOURS    = 24, // number of hours
3     parameter int N_MINUTES  = 60, // number of minutes
4     parameter int N_SECONDS  = 60, // number of seconds
5
6     // Output port widths
7     parameter int W_HOURS    = 5,
8     parameter int W_MINUTES  = 6,
9     parameter int W_SECONDS  = 6
10 ) (
11     input logic clk,
12     input logic enable,
13     output logic [W_HOURS-1:0] hours,
14     output logic [W_MINUTES-1:0] minutes,
15     output logic [W_SECONDS-1:0] seconds
16 );
```

When enable is low, all outputs retain their values.

When enable is high, seconds increments on each rising clock edge. On a seconds rollover (when seconds is at `N_SECONDS-1` and enable is high), seconds wraps to zero and minutes increments. Likewise, on a minutes rollover, minutes wraps to zero and hours increments; on an hours rollover, hours wraps to zero.

**Rationale** We introduced parameters for testing, while deliberately keeping the module name specific rather than generic. Smaller parameter values do more than speed up simulation: they make bugs easier to find, since the test bench is shorter and debugging is more manageable. Had we intended this module to be reused as a general cascaded counter, we would have given it a more generic name instead. Because its name is specific, it is fine to instantiate `hms_counter` without overriding any parameters: the defaults already implement exactly what the name promises.

**Implementation Notes** Follow these conventions in the implementation:

- Declare two **logic** variables named `second_rollover` and `minute_rollover`.
- Instantiate three `up_down_counter` modules called `u_hour`, `u_minute`, and `u_second`. The `u_` prefix, short for unit, marks these as module instances and avoids name clashes with signals.

To eliminate linter warnings, we must use width-adjusted parameter versions. The following contains a subtle bug:

```
1 // Incorrect!
2 localparam logic [W_MINUTES-1:0] NMinutes = W_MINUTES'(N_MINUTES);
3 localparam logic [W_SECONDS-1:0] NSeconds = W_SECONDS'(N_SECONDS);
```

The widths passed to the module are suitable for storing numbers from 0 to  $N - 1$ , but the code above stores  $N$  using the same number of bits. In edge cases, such as  $N = 2$ , it stores 0 instead of 2 due to truncation that you explicitly authorised.

Because the code needs  $N - 1$  rather than  $N$ , the solution is straightforward:

```
1 localparam logic [W_MINUTES-1:0] MaxMinutes = W_MINUTES'(N_MINUTES - 1);
2 localparam logic [W_SECONDS-1:0] MaxSeconds = W_SECONDS'(N_SECONDS - 1);
```

**Careful bit-width tracking is vital:** writing a logic condition like  $x == y+1$  is dangerous because  $y+1$  might wrap around to zero. This danger exists in other programming languages too, but it is most apparent in hardware design, where we work with small numbers of bits. The extra effort to be precise with widths costs minutes and can save hours of debugging.

For linting and compilation, include all required files.

```
verilator -Wall --lint-only rtl/hms_counter.sv rtl/up_down_counter.sv
iverilog -Wall -g2012 rtl/hms_counter.sv rtl/up_down_counter.sv tb/
    wave_hms_counter.v
```

```
rtl > hms_counter.sv > SystemVerilog - Language Support > hms_counter
1 // HMS Counter
2 // This module implements a Hours-Minutes-Seconds (HMS) counter using three
3 // up-down counters. The counter increments the seconds, minutes, and hours
4 // based on the enable signal and rolls over when reaching the maximum values.
5 //
6 // The hours, minutes and seconds are 'ticked up' by the up_down_counter
7 // modules which are instantiated for each time unit. The up_down_counter
8 // modules are the ones that handle the wrap back to 0 when the maximum value
9 // is reached, as determined by the MAX parameter argument that was passed.
10 // into it. Up is always up.
11 // The rollover is handled here, at the Next-State Logic section, which
12 // checks if the seconds or minutes have reached their maximum values and sets
13 // the corresponding rollover signals accordingly.
14 //
15 // Parameters:
16 //   N_HOURS:      - Number of hours (default: 24).
17 //   N_MINUTES:    - Number of minutes (default: 60).
18 //   N_SECONDS:    - Number of seconds (default: 60).
19 //   W_HOURS:      - Width of the hours output (default: 5).
20 //   W_MINUTES:    - Width of the minutes output (default: 6).
21 //   W_SECONDS:    - Width of the seconds output (default: 6).
22 //
23 // Ports:
24 //   clk:          - Clock input.
25 //   enable:       - When high, the counter is enabled.
26 //   hours [W_HOURS-1:0]: - Output for hours (0 to N_HOURS-1).
27 //   minutes [W_MINUTES-1:0]: - Output for minutes (0 to N_MINUTES-1).
28 //   seconds [W_SECONDS-1:0]: - Output for seconds (0 to N_SECONDS-1).
29 //
30
31 `timescale 1ns / 1ps
32
33 module hms_counter #(
34     parameter int N_HOURS = 24, // number of hours
35     parameter int N_MINUTES = 60, // number of minutes
36     parameter int N_SECONDS = 60, // number of seconds
37
38     // Output Port Widths
39     parameter int W_HOURS = 5, // bit length 5'b for hours (0-23)
40     parameter int W_MINUTES = 6, // bit length 6'b for minutes (0-59)
41     parameter int W_SECONDS = 6 // bit length 6'b for seconds (0-59)
42 ) (
43     input logic clk,
44     input logic enable,
45     output logic [W_HOURS-1:0] hours, // = W_HOURS'(0),
46     output logic [W_MINUTES-1:0] minutes, // = W_MINUTES'(0),
47     output logic [W_SECONDS-1:0] seconds // = W_SECONDS'(0)
48 );
49
```

```

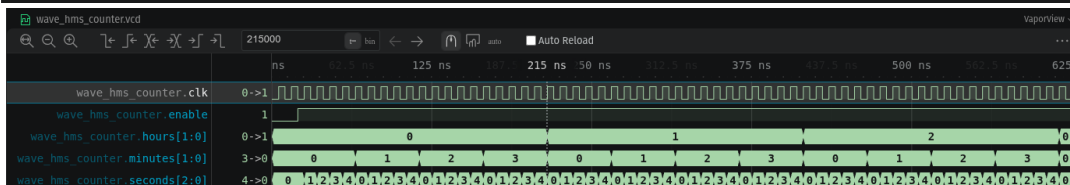
rtl > @ hms_counter.sv > ...
47     output logic [W_SECONDS-1:0] seconds // = W_SECONDS'(0)
48 );
49
50 // Declare Local Parameters and Signals
51 localparam logic [W_MINUTES-1:0] MaxMinutes = W_MINUTES'(N_MINUTES - 1);
52 localparam logic [W_SECONDS-1:0] MaxSeconds = W_SECONDS'(N_SECONDS - 1);
53 logic second_rollover, minute_rollover;
54
55 // State Logic
56 up_down_counter #(
57     .MAX (N_HOURS - 1),
58     .WIDTH(W_HOURS)
59 ) u_hour (
60     .clk(clk),
61     .enable(minute_rollover),
62     .up(1'b1),
63     .count(hours)
64 );
65
66 up_down_counter #(
67     .MAX (N_MINUTES - 1),
68     .WIDTH(W_MINUTES)
69 ) u_minute (
70     .clk(clk),
71     .enable(second_rollover),
72     .up(1'b1),
73     .count(minutes)
74 );
75
76 up_down_counter #(
77     .MAX (N_SECONDS - 1),
78     .WIDTH(W_SECONDS)
79 ) u_second (
80     .clk(clk),
81     .enable(enable),
82     .up(1'b1),
83     .count(seconds)
84 );
85
86 // Next-State Logic
87 always_comb begin
88     if (minutes == MaxMinutes && seconds == MaxSeconds) minute_rollover = 1'b1;
89     else minute_rollover = 1'b0;
90     if (seconds == MaxSeconds) second_rollover = 1'b1;
91     else second_rollover = 1'b0;
92 end
93
94 endmodule
95

```

```

• root@4034a70d84cc:/workspace# verilator -Wall --lint-only rtl/hms_counter.sv rtl/up_down_counter.sv
• root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/hms_counter.sv rtl/up_down_counter.sv
• root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/hms_counter.sv rtl/up_down_counter.sv tb/wave_hms_counter.v
• root@4034a70d84cc:/workspace# ./a.out
VCD info: dumpfile wave_hms_counter.vcd opened for output.
tb/wave_hms_counter.v:39: $finish called at 620000 (1ps)
❖ root@4034a70d84cc:/workspace#

```



```

• root@4034a70d84cc:/workspace# pytest tests/test_hms_counter.py
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0
rootdir: /workspace
configfile: .pytest.toml
plugins: metadata-3.1.1, html-4.2.0
collected 3 items

tests/test_hms_counter.py ... [100%]

===== 3 passed in 0.97s =====
❖ root@4034a70d84cc:/workspace#

```

## 6.5. Modulo N Counter

---

```
1 module mod_n_counter #(
2     parameter int N = 4,
3     parameter int WIDTH = 2
4 ) (
5     input logic clk,
6     input logic rst,
7     input logic enable,
8     output logic [WIDTH-1:0] count
9 );
```

---

The output `count` is initially zero.

On a rising clock edge, `rst` takes priority over `enable`: a high `rst` clears `count` regardless of `enable`.

If only `enable` is high, `count` advances. When it reaches  $N - 1$ , the next enabled edge wraps it back to zero; otherwise it increments by one.

**Rationale** Even if `up_down_counter` had a reset pin, a dedicated module with a minimal interface makes the design intent clearer than one with several pins tied high or low.

**Implementation Notes** Refer to your previous modules for the techniques needed to suppress linter warnings.

It is customary<sup>2</sup> to place reset logic inside the flip-flop block rather than in the next-state logic. Use `always_ff @(posedge clk)` rather than `always @(posedge clk)`; the linter will not flag the latter, but `always_ff` makes the intent explicit.

---

```
1 always_ff @(posedge clk)
2     if (rst) count <= '0;
3     else if (enable) count <= next_count;
```

---

---

<sup>2</sup>This mirrors real flip-flops, many of which have a dedicated reset pin.

```

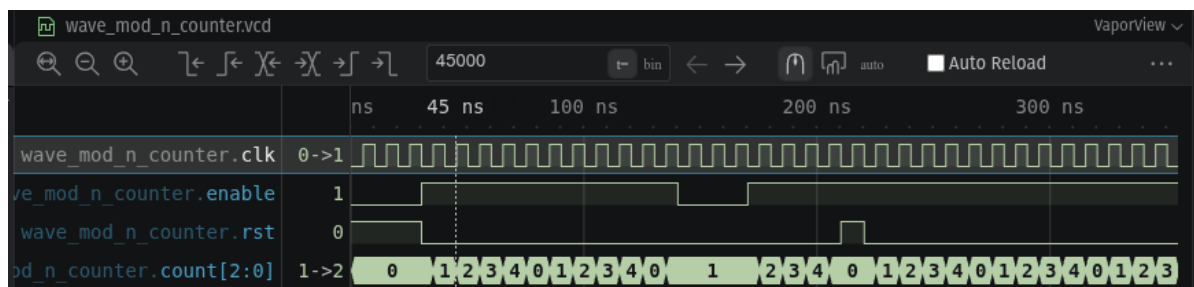
rtl > @ mod_n_counter.sv > ...
1 // Modulo N Counter
2 // This module implements a modulo N counter that counts from 0 to N-1 and wraps
3 // around to 0 when it reaches N. The counter can be reset asynchronously and is
4 // enabled by an input signal.
5 //
6 // Parameters:
7 // N - Maximum count value (default: 4).
8 // WIDTH - Width of the count output (default: 2).
9 //
10 // Ports:
11 // clk - Clock input.
12 // rst - Asynchronous reset input.
13 // enable - When high, the counter is enabled.
14 // count - Output count value (0 to N-1).
15 //
16
17 `timescale 1ns / 1ps
18
19 module mod_n_counter #(
20     parameter int N = 4,
21     parameter int WIDTH = 2
22 ) (
23     input logic clk,
24     input logic rst,
25     input logic enable,
26     output logic [WIDTH-1:0] count = WIDTH'(0)
27 );
28
29 // Declare local parameters and signals
30 // Cast input `N` parameter to local `Max` parameter
31 localparam logic [WIDTH-1:0] Max = WIDTH'(N - 1);
32 logic [WIDTH-1:0] next_count;
33
34 // State, and Output Logic
35 always_ff @(posedge clk) begin
36     if (rst) count <= WIDTH'(0); // Asynchronous reset to 0
37     else if (enable) count <= next_count;
38 end
39
40 // Next-State Logic
41 always_comb begin
42     if (count < Max) next_count = count + WIDTH'(1);
43     else next_count = WIDTH'(0); // Wrap to 0 when Max
44 end
45
46 endmodule
47

```

```

root@4034a70d84cc:/workspace# verilator -Wall --lint-only rtl/mod_n_counter.sv
root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/mod_n_counter.sv
root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/mod_n_counter.sv tb/wave_mod_n_counter.v
root@4034a70d84cc:/workspace# ./a.out
VCD info: dumpfile wave_mod_n_counter.vcd opened for output.
tb/wave_mod_n_counter.v:47: $finish called at 350000 (1ps)
❖ root@4034a70d84cc:/workspace#

```



```

root@4034a70d84cc:/workspace# pytest tests/test_mod_n_counter.py
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0
rootdir: /workspace
configfile: .pytest.toml
plugins: metadata-3.1.1, html-4.2.0
collected 2 items

tests/test_mod_n_counter.py .. [100%]

===== 2 passed in 0.65s =====
❖ root@4034a70d84cc:/workspace#

```

## 6.6. Restartable Rate Generator

```
1 module restartable_rate_generator #(
2     parameter int CYCLE_COUNT = 2
3 ) (
4     input logic clk,
5     input logic run,
6     output logic tick
7 );
```

When run is low, tick goes low.

When run is high:

- After run has been high for exactly `CYCLE_COUNT-1` rising edges, tick is high for exactly one clock cycle.
- Thereafter, after each period of run having remained high for `CYCLE_COUNT` rising edges, tick is again high for exactly one clock cycle.

The edge case `CYCLE_COUNT = 1` is supported: tick follows run directly.

The module is to be implemented as a Moore FSM.

**Rationale** This module drives the watch's one-second timekeeping and is reused during time editing, where holding a button causes repeated increments or decrements. Supporting `CYCLE_COUNT = 1` ensures the design remains functional when the watch is configured to tick every clock cycle, which is useful for debugging.

There is no functional reason for this module to be a Moore FSM; the requirement is imposed purely as a learning opportunity.

**Implementation Notes** There is no elegant way to write this module directly: it is just complex enough that a direct implementation becomes awkward. Instantiating a `mod_n_counter` avoids this, keeping the higher-level logic uncluttered.

In a Moore FSM, outputs depend only on state. Since tick must be low whenever run is sampled low, run must be captured as part of the state.

For `CYCLE_COUNT > 1`, the remaining state is a counter with modulus `CYCLE_COUNT`. Its width is an internal concern — it does not affect any port — and must be computed internally rather than imposed on the instantiating module.

```
1 localparam int CounterWidth = $clog2(CYCLE_COUNT);
```

### Report 5

Explain how `localparam CounterWidth = $clog2(CYCLE_COUNT);` works.

`CYCLE_COUNT` is a (unsigned) integer of width 32  
`$clog2(number)` returns the number of bits required cover the number in a binary string

When `CYCLE_COUNT = 1`, `CounterWidth` is zero, so no counter is needed; a `generate` statement handles this by conditionally omitting the instantiation. Blocks inside a `generate` statement must be named — here `g_general` and `g_special` — or the linter will warn.

```
1 generate
2     if (CYCLE_COUNT > 1) begin : g_general
3         // Put code for the general case here,
4         // including counter instantiation
5     end else begin : g_special
6         // Put code for the special case here
7     end
8 endgenerate
```

With readability in mind, we start with the high-level structure:

```
1 // Becomes high at the end of each cycle
2 logic tick_qualifier;
3
4 logic running = 1'b0;
5 always_ff @(posedge clk) running <= run;
6
7 assign tick = running && tick_qualifier;
```

It remains for the `generate` block to provide `tick_qualifier`. The special case is straightforward:

```
1 begin : g_special
2     assign tick_qualifier = 1'b1;
3 end
```

A template for the general case follows, with the logic kept separate from the module instantiation.

```
1 begin : g_general
2     localparam int CountWidth = $clog2(CYCLE_COUNT);
3
4     logic rst_count;
5     logic enable_count;
6     logic [CountWidth-1:0] count;
7     mod_n_counter #(
8         .N(CYCLE_COUNT),
9         .WIDTH(CountWidth)
10    ) u_count (
11        .clk(clk),
12        .rst(rst_count),
13        .enable(enable_count),
14        .count(count)
15    );
16
17     assign rst_count = ...; // Fill in
18     assign enable_count = ...; // Fill in
19
20     assign tick_qualifier = ...; // Fill in
21 end
```

To ensure the correct bit width, use `CountWidth'(CYCLE_COUNT - 1)` when needed.

#### Report 6

In addition to running the provided test bench, write your own test bench verifying correct behaviour when `CYCLE_COUNT = 1`.

**Note** The module's behaviour may appear asymmetric: the initial delay is `CYCLE_COUNT-1` rising edges, whereas subsequent ticks repeat every `CYCLE_COUNT` rising edges. However, if run is synchronised to the clock (i.e., it transitions just after a rising edge, as it would if driven from a flip-flop), the initial delay is also `CYCLE_COUNT` clock cycles, matching the repeat rate.

```

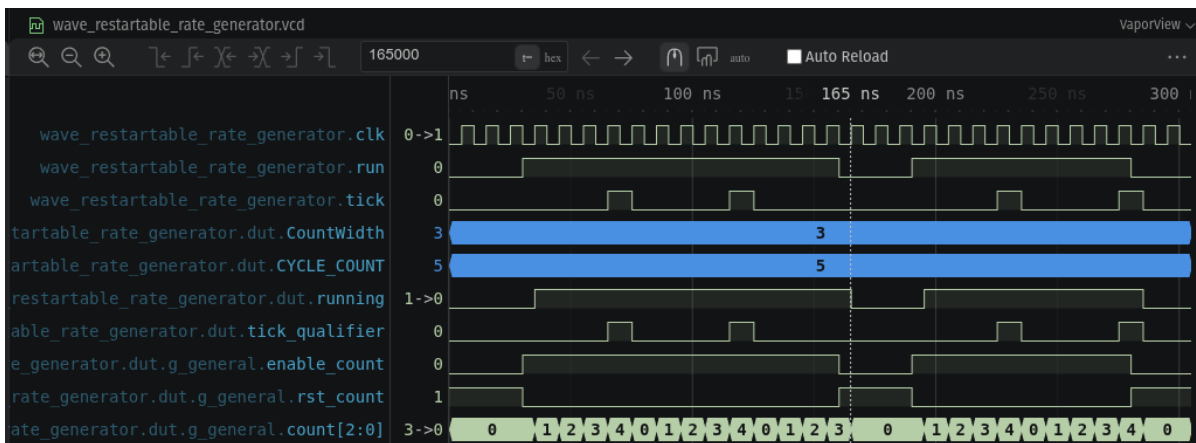
rtl > restartable_rate_generator.sv > SystemVerilog - Language Support > restartable_rate_generator
1 // Restartable Rate Generator
2 // Generates a Mod-N counter that produces a tick at a specified rate when
3 // enabled. The rate is defined by the CYCLE_COUNT parameter. The generated
4 // Mod_CYCLE_COUNT counter can be restarted by deasserting and reasserting
5 // the run signal.
6 //
7 // Parameters:
8 //   CYCLE_COUNT: - Number of clock cycles per tick (default: 2).
9 //
10 // Ports:
11 //   clk:         - Clock input.
12 //   run:         - When high, the generated Mod_CYCLE_COUNT instant is
13 //                 enabled and produces ticks after successfully reaching
14 //                 each full cycle.
15 //   tick:        - Output tick signal, high when both run is high and the
16 //                 tick_qualifier is high.
17 //
18 // Local Variables:
19 //   CountWidth:  - Bit width of parameter CYCLE_COUNT.
20 //   rst_count:   - Reset signal for the generated `mod_n_counter` counter.
21 //   enable_count: - Enable signal for the generated `mod_n_counter` counter.
22 //   count:       - Count output of the generated `mod_n_counter` counter.
23 //   tick_qualifier: - High when the generated `mod_n_counter` counter reaches
24 //                     its maximum value.
25 //   running:     - Indicates whether the rate generator is enabled based
26 //                 on input signal `run`.
27
28 `timescale 1ns / 1ps
29
30 module restartable_rate_generator #(
31     parameter int CYCLE_COUNT = 2
32 ) (
33     input logic clk,
34     input logic run,
35     output logic tick
36 );
37
38 // Declare Local Parameters and Signals
39 // Get required bit width for the counter based on CYCLE_COUNT
40 localparam int CountWidth = $clog2(CYCLE_COUNT);
41
42 // Becomes high at the end of each cycle
43 logic tick_qualifier;
44
45 generate
46     if (CYCLE_COUNT > 1) begin : g_general
47         logic rst_count;
48         logic enable_count;
49         logic [CountWidth-1:0] count;
50
51         .N(CYCLE_COUNT),
52         .WIDTH(CountWidth)
53     ) u_count (
54         .clk(clk),
55         .rst(rst_count),
56         .enable(enable_count),
57         .count(count)
58     );
59
60     assign rst_count = !run;
61     assign enable_count = run;
62     assign tick_qualifier = (count == CountWidth'(CYCLE_COUNT - 1));
63 end else begin : g_special
64     assign tick_qualifier = 1'b1;
65 end
66 endgenerate
67
68 // Running state logic
69 logic running = 1'b0;
70 always_ff @(posedge clk) running <= run;
71
72 // Output Logic
73 assign tick = running && tick_qualifier;
74
75 endmodule

```

```

root@4034a70d84cc:/workspace# verilog -Wall --lint-only rtl/restartable_rate_generator.vhdl rtl/mod_n_counter.vhdl
root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/restartable_rate_generator.vhdl rtl/mod_n_counter.vhdl
root@4034a70d84cc:/workspace# iverilog -Wall -g2012 rtl/restartable_rate_generator.vhdl rtl/mod_n_counter.vhdl tb/wave_restartable_rate_generator.v
root@4034a70d84cc:/workspace# ./a.out
VCD info: dumpfile wave_restartable_rate_generator.vcd opened for output.
tb/wave_restartable_rate_generator.v:33: $finish called at 300000 (1ps)
root@4034a70d84cc:/workspace#

```



```

(venv) root@4034a70d84cc:/workspace# pytest tests/test_restartable_rate_generator.py
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0
rootdir: /workspace
configfile: .pytest.toml
plugins: metadata-3.1.1, html-4.2.0
collected 6 items

tests/test_restartable_rate_generator.py ..... [100%]

===== 6 passed in 1.77s =====
(venv) root@4034a70d84cc:/workspace#

```

## 6.7. Variable-Speed Time Display

```
1 module top_time_display_v1 #(
2     parameter int CYCLES_PER_SECOND = 50_000_000
3 ) (
4     input logic CLOCK_50,
5     input logic [1:0] SW,
6     output logic [6:0] HEX5,
7     output logic [6:0] HEX4,
8     output logic [6:0] HEX3,
9     output logic [6:0] HEX2,
10    output logic [6:0] HEX1,
11    output logic [6:0] HEX0
12 );
```

| SW[1:0] | Tick Rate |
|---------|-----------|
| 2'b00   | 1 Hz      |
| 2'b01   | 25 Hz     |
| 2'b10   | 1 kHz     |
| 2'b11   | 50 MHz    |

This top-level module for the DE1-SoC board displays the time on the seven-segment displays, initialised to 00:00:00, with the tick rate controlled by SW[1:0].

**Rationale** The goal is to have a working design ready for the DE1-SoC board when you arrive at the lab. Designing and debugging in VS Code beforehand means it should work first time.

**Implementation Notes** The final code may be around 100 lines but is straightforward. Begin by instantiating:

- one `hms_counter`;
- three `restartable_rate_generators`, one for each of the slower tick rates;
- three `binary_to_bcds`, one each for hours, minutes and seconds; and
- six `seven_segments`, one for each of the digits.

If it is not immediately obvious what to connect to a port, create a named `logic` variable for it and assign it a value using an `assign` statement at the bottom of the module. This keeps instantiation separate from logic and improves readability, provided the variables are named clearly.

The outputs of `hms_counter` are narrower than the bin input of `binary_to_bcd`. Explicitly zero-extend them when making each connection, for example `.bin({2'b0, hours})`.

For each `restartable_rate_generator`, use `CYCLES_PER_SECOND` to derive the `CYCLE_COUNT` value. The default of `50_000_000` matches the DE1-SoC's 50 MHz clock; the test suite overrides it to a smaller value to keep simulation fast.

As the design grows, we can save time by telling `verilator` or `iverilog` to search `rtl/` for any modules it needs. For `iverilog`, we also specify `.sv` as the file extension to search. Note that `-Y .sv` must come before `-y rtl`.

```
verilator -Wall --lint-only -y rtl rtl/top_time_display_v1.sv
iverilog -Wall -g2012 -Y .sv -y rtl tb/wave_top_time_display_v1.v
```

The automated tests check the 1 kHz tick rate but not the slower rates (25 Hz and 1 Hz); these can be verified by hand once the design is on the board.

The simulator may produce warnings labelled Time: 0, such as the following.

```
root@a374924768bc:/workspace# ./a.out
VCD info: dumpfile wave_top_time_display_v1.vcd opened for output.
WARNING: rtl/seven_segment.sv:30: value is unhandled for priority or
        unique case statement
        Time: 0 Scope: wave_top_time_display_v1.dut.u_HEX0
```

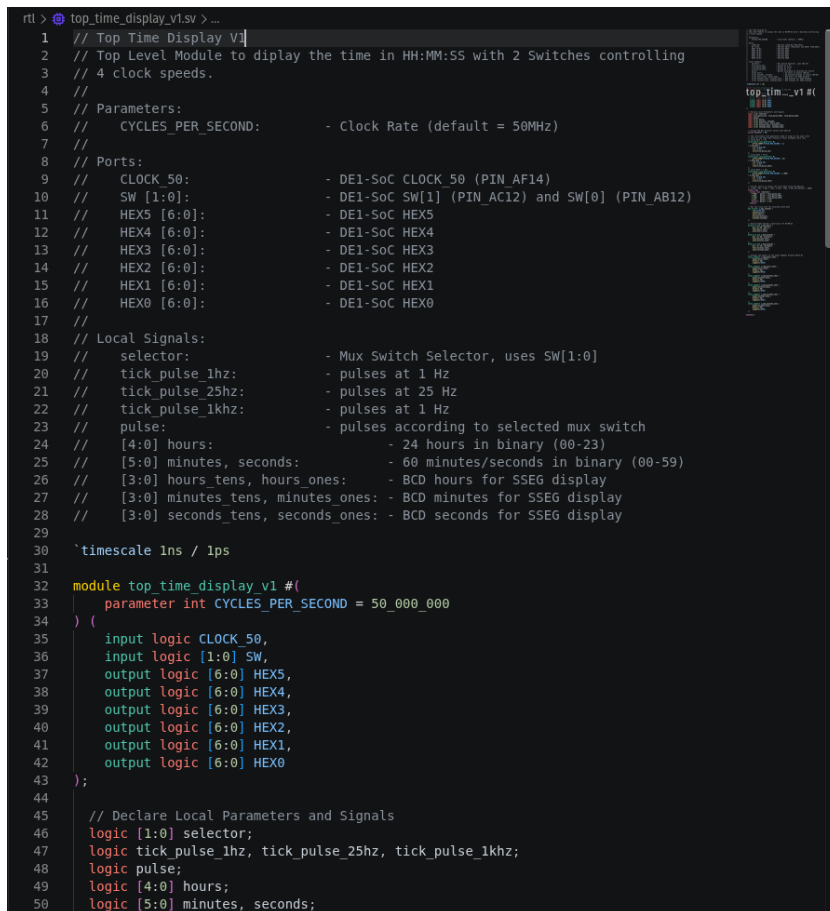
This is due to a race condition: the **unique case** statement in `seven_segment` must have executed before at least one of the `up_down_counter` instances has initialised its output, so some or all bits of `digit` are unknown. Since no case item matches an unknown value, the **unique** constraint is correctly reported as violated. There are three solutions.

1. Ignore **unique case** warnings labelled Time: 0.
2. Add a `$isunknown(digit)` check to handle unknown inputs explicitly.
3. Some simulators allow warnings at time 0 to be suppressed.

If these warnings truly bother you, add the following to `seven_segment`:

```
if (blank || $isunknown(digit)) segments_active_high = '0';
```

Quartus will ignore this construct, and since the warnings are a simulator quirk, adding extra logic to suppress them is not ideal. The most pragmatic solution is simply to ignore **unique case** warnings labelled Time: 0.



```
rtl > top_time_display_v1.sv >...
1 // Top Time Display V1
2 // Top Level Module to display the time in HH:MM:SS with 2 Switches controlling
3 // 4 clock speeds.
4 //
5 // Parameters:
6 // CYCLES_PER_SECOND: - Clock Rate (default = 50MHz)
7 //
8 // Ports:
9 // CLOCK_50: - DE1-SoC CLOCK 50 (PIN AF14)
10 // SW [1:0]: - DE1-SoC SW[1] (PIN_AC12) and SW[0] (PIN_AB12)
11 // HEX5 [6:0]: - DE1-SoC HEX5
12 // HEX4 [6:0]: - DE1-SoC HEX4
13 // HEX3 [6:0]: - DE1-SoC HEX3
14 // HEX2 [6:0]: - DE1-SoC HEX2
15 // HEX1 [6:0]: - DE1-SoC HEX1
16 // HEX0 [6:0]: - DE1-SoC HEX0
17 //
18 // Local Signals:
19 // selector: - Mux Switch Selector, uses SW[1:0]
20 // tick_pulse_1hz: - pulses at 1 Hz
21 // tick_pulse_25hz: - pulses at 25 Hz
22 // tick_pulse_1khz: - pulses at 1 kHz
23 // pulse: - pulses according to selected mux switch
24 // [4:0] hours: - 24 hours in binary (00-23)
25 // [5:0] minutes, seconds: - 60 minutes/seconds in binary (00-59)
26 // [3:0] hours_tens, hours_ones: - BCD hours for SSEG display
27 // [3:0] minutes_tens, minutes_ones: - BCD minutes for SSEG display
28 // [3:0] seconds_tens, seconds_ones: - BCD seconds for SSEG display
29
30 `timescale 1ns / 1ps
31
32 module top_time_display_v1 #(
33     parameter int CYCLES_PER_SECOND = 50_000_000
34 ) (
35     input logic CLOCK_50,
36     input logic [1:0] SW,
37     output logic [6:0] HEX5,
38     output logic [6:0] HEX4,
39     output logic [6:0] HEX3,
40     output logic [6:0] HEX2,
41     output logic [6:0] HEX1,
42     output logic [6:0] HEX0
43 );
44
45 // Declare Local Parameters and Signals
46 logic [1:0] selector;
47 logic tick_pulse_1hz, tick_pulse_25hz, tick_pulse_1khz;
48 logic pulse;
49 logic [4:0] hours;
50 logic [5:0] minutes, seconds;
```

rtl > top\_time\_display\_v1.sv > SystemVerilog - Language Support > top\_time\_display\_v1 > {} seconds

```
50 logic [5:0] minutes, seconds;
51 logic [3:0] hours_tens, hours_ones;
52 logic [3:0] minutes_tens, minutes_ones;
53 logic [3:0] seconds_tens, seconds_ones;
54
55 // Assign the Mux Selector switch with SW[1:0]
56 assign selector = SW;
57
58 // The restartable rate generators need to align to the same clock
59 // while at the same time ticking to their assigned clock rate
60 // Clock Rate = 1 Hz
61 restartable_rate_generator #(
62     .CYCLE_COUNT(CYCLES_PER_SECOND / 1)
63 ) u_1hz_gen (
64     .clk (CLOCK_50),
65     .run (1'b1),
66     .tick(tick_pulse_1hz)
67 );
68 // Clock Rate = 25 Hz
69 restartable_rate_generator #(
70     .CYCLE_COUNT(CYCLES_PER_SECOND / 25)
71 ) u_25hz_gen (
72     .clk (CLOCK_50),
73     .run (1'b1),
74     .tick(tick_pulse_25hz)
75 );
76 // Clock Rate = 1 kHz
77 restartable_rate_generator #(
78     .CYCLE_COUNT(CYCLES_PER_SECOND / 1_000)
79 ) u_1khz_gen (
80     .clk (CLOCK_50),
81     .run (1'b1),
82     .tick(tick_pulse_1khz)
83 );
84
85 // The Mux Switch to select the Clock Rate using the SW[1:0]
86 // 2'b00 = 1Hz, 2'b01 = 25Hz, 2'b10 = 1kHz, 2'b11 and default = 50MHz
87 always comb
88     unique case (selector)
89         2'b00: pulse = tick_pulse_1hz;
90         2'b01: pulse = tick_pulse_25hz;
91         2'b10: pulse = tick_pulse_1khz;
92         2'b11: pulse = 1'b1;
93         default: pulse = 1'b1;
94     endcase
95
96 // Run the clock with the selected Clock Rate
97 hms_counter u_hms_counter (
98     .clk(CLOCK_50),
99     .enable(pulse),
```

rtl > top\_time\_display\_v1.sv > SystemVerilog - Language Support > top\_time\_display\_v1 > case

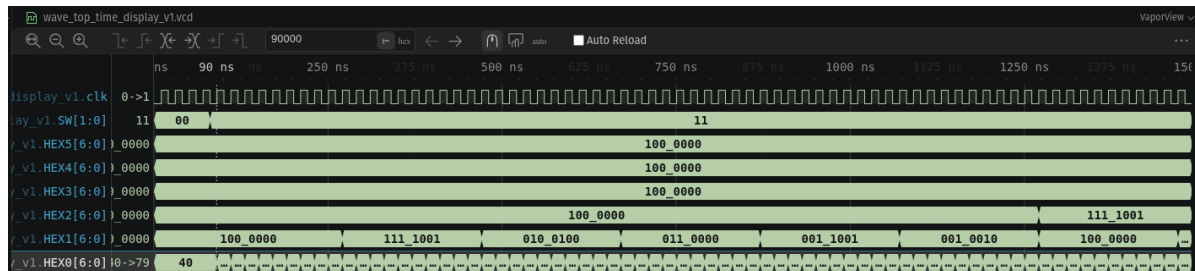
```
98     .clk(CLOCK_50),
99     .enable(pulse),
100     .hours(hours),
101     .minutes(minutes),
102     .seconds(seconds)
103 );
104
105 // Binary-Coded Decimal conversions for HH:MM:SS
106 binary_to_bcd u_bcd_hours (
107     .bin ({2'b0, hours}),
108     .tens(hours_tens),
109     .ones(hours_ones)
110 );
111 binary_to_bcd u_bcd_minutes (
112     .bin ({1'b0, minutes}),
113     .tens(minutes_tens),
114     .ones(minutes_ones)
115 );
116 binary_to_bcd u_bcd_seconds (
117     .bin ({1'b0, seconds}),
118     .tens(seconds_tens),
119     .ones(seconds_ones)
120 );
121
122 // Display the digits on the Seven Segment Display HEX[5:0]
123 seven_segment u_seg_hours_tens (
124     .digit(hours_tens),
125     .blank(1'b0),
126     .segments(HEX5)
127 );
128 seven_segment u_seg_hours_ones (
129     .digit(hours_ones),
130     .blank(1'b0),
131     .segments(HEX4)
132 );
133 seven_segment u_seg_minutes_tens (
134     .digit(minutes_tens),
135     .blank(1'b0),
136     .segments(HEX3)
137 );
138 seven_segment u_seg_minutes_ones (
139     .digit(minutes_ones),
140     .blank(1'b0),
141     .segments(HEX2)
142 );
143 seven_segment u_seg_seconds_tens (
144     .digit(seconds_tens),
145     .blank(1'b0),
146     .segments(HEX1)
147 );
```

```
143 seven_segment u_seg_seconds_tens (
144     .digit(seconds_tens),
145     .blank(1'b0),
146     .segments(HEX1)
147 );
148 seven_segment u_seg_seconds_ones (
149     .digit(seconds_ones),
150     .blank(1'b0),
151     .segments(HEX0)
152 );
153
154 endmodule
155
```

```

• (venv) root@4034a70d84cc:/workspace# verilator -Wall --lint-only -y rtl rtl/top_time_display_v1.sv
• (venv) root@4034a70d84cc:/workspace# iverilog -Wall -g2012 -Y .sv -y rtl rtl/top_time_display_v1.sv
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/top_time_display_v1.sv:88: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
• (venv) root@4034a70d84cc:/workspace# iverilog -Wall -g2012 -Y .sv -y rtl rtl/top_time_display_v1.sv tb/wave_top_time_display_v1.v
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/seven_segment.sv:28: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
rtl/top_time_display_v1.sv:88: vvp.tgt sorry: Case unique/unique0 qualities are ignored.
• (venv) root@4034a70d84cc:/workspace# ./a.out
VCD info: dumpfile wave_top_time_display_v1.vcd opened for output.
tb/wave_top_time_display_v1.v:34: $finish called at 1480000 (1ps)
❖ (venv) root@4034a70d84cc:/workspace#

```



```

• (venv) root@4034a70d84cc:/workspace# pytest tests/test_top_time_display_v1.py
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0
rootdir: /workspace
configfile: .pytest.toml
plugins: metadata-3.1.1, html-4.2.0
collected 1 item

tests/test_top_time_display_v1.py . [100%]

===== 1 passed in 0.30s =====
❖ (venv) root@4034a70d84cc:/workspace#

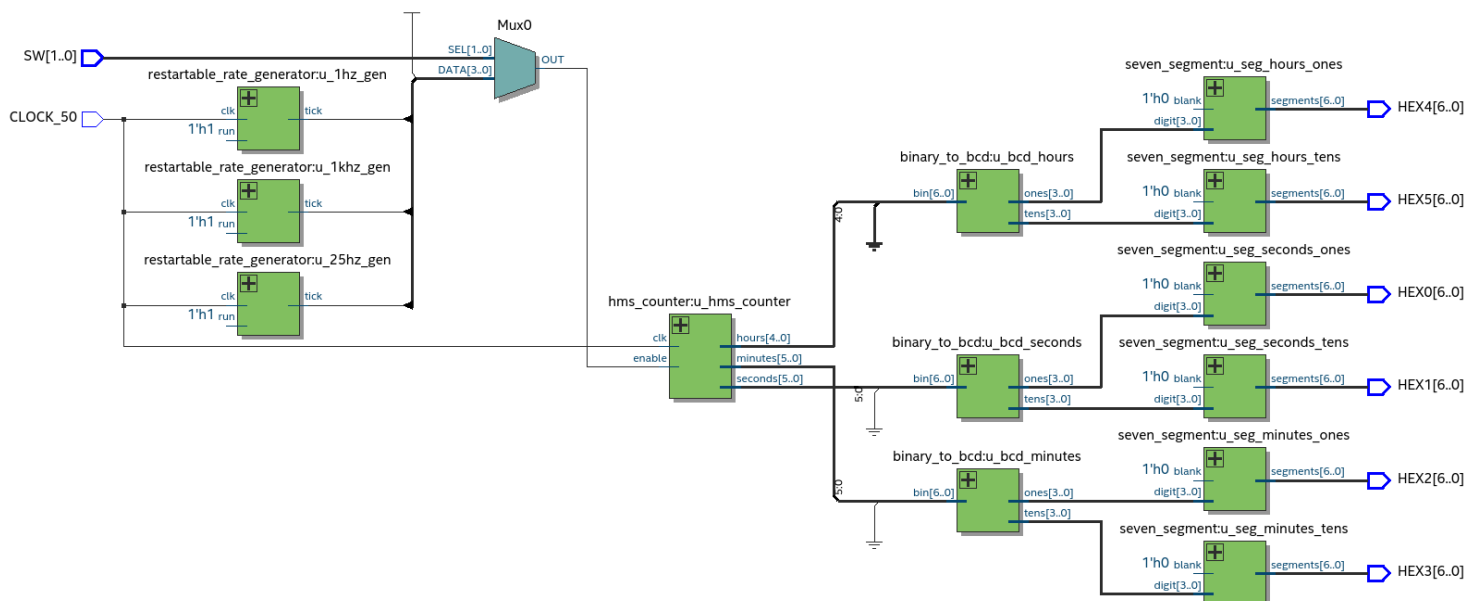
```

## 6.8. Programming the Board

Skip to the next section if you are not in the lab; return here when you are ready to program the board.

Follow these steps to program the DE1-SoC:

1. Create a new Quartus project (e.g., `digital_watch`) with the correct settings for the DE1-SoC board.
2. Set the top-level module to `top_time_display_v1`.
3. Import assignments from `DE1_SoC.qsf`.
4. Create an `.SDC` file specifying the clock period.
5. Copy all your `rtl/*.sv` files into the Quartus project root directory (or under `rtl/` if you prefer).
6. Go to **Project** > **Add/Remove Files** and add these files to the project.
7. Compile and program the board.



## 7. Settable Watch

By the end of this section, we will have a settable watch: time displayed on the DE1-SoC board, incrementing once per second, with the ability to edit the time using the keys. Holding KEY[3] for at least one second enters edit mode, with the seconds digits flashing at 2 Hz with an 80% duty cycle. A second press of KEY[3] moves to the minutes digits, a third to the hours digits, and a fourth exits edit mode.

In edit mode, KEY[1] increments and KEY[0] decrements the flashing digits. Holding either key for more than half a second repeats the increment or decrement at 10 Hz.

### 7.1. Fixed-Frequency Fixed-Duty PWM Generator

```
1 module pwm_generator #(
2     // Number of clock cycles in one PWM period
3     parameter int PERIOD_CYCLES = 50_000_000,
4
5     // Number of clock cycles output is high
6     parameter int DUTY_CYCLES = 25_000_000
7 ) (
8     input logic clk,
9     input logic rst,
10    output logic pwm_out
11 );
```

The output `pwm_out` is periodic with a period of `PERIOD_CYCLES` clock cycles, high for the first `DUTY_CYCLES` cycles of each period and low for the remainder. When `rst` is asserted, the output restarts from the next rising clock edge.

#### Report 7

Explain why this module is called a PWM generator.

**Rationale** During edit mode, the display flashes to indicate which field is being edited. This module provides the PWM signal that drives the flashing.

**Implementation Notes** Instantiate a `mod_n_counter`. To see its port names while working, right-click `mod_n_counter.sv` in the Explorer panel and select *Open to the Side*.

#### Report 8

In `mod_n_counter`, `rst` takes priority over `enable`. Comment on the benefit of this design choice.

[https://en.wikipedia.org/wiki/Pulse-width\\_modulation](https://en.wikipedia.org/wiki/Pulse-width_modulation)

<https://www.sameskydevices.com/blog/pulse-width-modulation-pwm-how-it-works-and-why-its-essential-in-electronics>

**Pulse-width modulation (PWM)**, also known as **pulse-duration modulation (PDM)**<sup>[a]</sup> or **pulse-length modulation (PLM)**,<sup>[1]</sup> is any method of representing a signal as a [rectangular wave](#) with a varying [duty cycle](#) (and for some methods also a varying [period](#)).

PWM facilitates the portion of the period for which the digital signal is active

## Summary

Modulation is the application of control over a device or system. Pulse width modulation in particular, is an effective and efficient method used to control power in a wide variety of electronic devices. It allows for precise control and low-power operation when compared to other control technologies and is used across many applications to exert analog-like control using a digital signal.

```
rtl >  pwm_generator.sv > ...
```

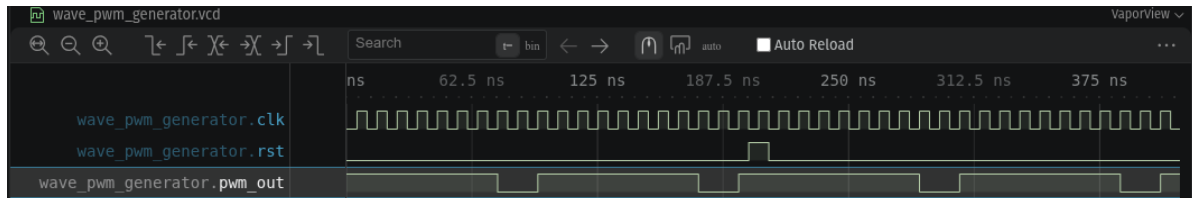
```
1 // Pulse Width Modulation (PWM) Generator
2 // Generates a PWM signal with a specified duty and period cycle. `pwm_out`
3 // signal is high for `DUTY_CYCLES` clock cycles and low for the remainder
4 // in `PERIOD_CYCLES` clock cycles. Local `count` signal generated by
5 // Modulo N Counter
6 //
7 // Parameters:
8 //   PERIOD_CYCLES: - Number of clock cycles in one PWM period
9 //                   (default: 50_000_000)
10 //   DUTY_CYCLES:  - Number of clock cycles output is high
11 //                   (default: 25_000_000)
12 //
13 // Ports:
14 //   clk:          - Clock input
15 //   rst:          - Asynchronous reset input
16 //   pwm_out:      - PWM output signal, high when within duty cycle,
17 //                   low when outside duty cycle
18 //
19 // Local Variables:
20 //   CountWidth:   - Bit width of parameter PERIOD_CYCLES.
21
22 `timescale 1ns / 1ps
23
24 module pwm_generator #(
25     // Number of clock cycles in one PWM period
26     parameter int PERIOD_CYCLES = 50_000_000,
27
28     // Number of clock cycles output is high
29     parameter int DUTY_CYCLES = 25_000_000
30 ) (
31     input logic clk,
32     input logic rst,
33     output logic pwm_out
34 );
35
36 // Declare local parameters and signals
37 // (+1'b1) is needed here to cover edge cases when PERIOD_CYCLES are powers
38 // of 2, otherwise there will be integer overflow when DUTY_CYCLES is equal
39 // to PERIOD_CYCLES where localparam `DutyCycles` will equal to 0
40 localparam int CountWidth = $clog2(PERIOD_CYCLES) + 1'b1;
41 localparam logic [CountWidth-1:0] DutyCycles = CountWidth'(DUTY_CYCLES);
42 logic [CountWidth-1:0] count;
43
44 // Instantiate Modulo N Counter with N = PERIOD_CYCLES.
45 mod_n_counter #(
46     .N(PERIOD_CYCLES),
47     .WIDTH(CountWidth)
48 ) u_period_cycle_count (
49     .clk(clk),
50     .rst(rst),
51     .enable(1'b1),
52     .count(count)
53 );
54
55 // `pwm_out` is high as for period of DutyCycles within PeriodCycles,
56 // past period of DutyCycles, `pwm_out` is low for rest of PeriodCycles.
57 always_comb begin
58     if (count < DutyCycles) pwm_out = 1'b1;
59     else pwm_out = 1'b0;
60 end
61
62 endmodule
63
```



```

• (venv) root@4034a70d84cc:/workspace# ./a.out
VCD info: dumpfile user_wave_pwm_generator.vcd opened for output.
tb/user_wave_pwm_generator.v:29: $finish called at 440000 (1ps)
• (venv) root@4034a70d84cc:/workspace# verilator -Wall -y rtl --lint-only rtl/pwm_generator.sv
• (venv) root@4034a70d84cc:/workspace# iverilog -Wall -g2012 -Y .sv -y rtl rtl/pwm_generator.sv
• (venv) root@4034a70d84cc:/workspace# iverilog -Wall -g2012 -Y .sv -y rtl tb/wave_pwm_generator.v
• (venv) root@4034a70d84cc:/workspace# ./a.out
VCD info: dumpfile wave_pwm_generator.vcd opened for output.
tb/wave_pwm_generator.v:29: $finish called at 410000 (1ps)
❖ (venv) root@4034a70d84cc:/workspace#

```



```

• (venv) root@4034a70d84cc:/workspace# pytest tests/test_pwm_generator.py
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0
rootdir: /workspace
configfile: .pytest.toml
plugins: metadata-3.1.1, html-4.2.0
collected 7 items

tests/test_pwm_generator.py ..... [100%]

===== 7 passed in 2.15s =====
❖ (venv) root@4034a70d84cc:/workspace#

```

## 7.2. Rising-Edge Detector

```
1 module rising_edge_detector (  
2     input  logic clk,  
3     input  logic sig_in,  
4     output logic rise  
5 );
```

The output `rise` is asserted immediately when `sig_in` transitions from low to high, and deasserted as soon as `sig_in` returns low or the next rising clock edge captures the high value, whichever comes first.

**Rationale** At 50 MHz, a button press spans millions of clock cycles. Without edge detection, a single press of the edit button could trigger millions of mode transitions. A rising edge detector converts the press into a single-cycle pulse, ensuring exactly one transition per press.

Unlike the other modules in this assignment, `rising_edge_detector` is a Mealy FSM — the industry-standard choice for edge detection — as it avoids a one-cycle output delay, is simpler to write, and produces output logic simple enough to be glitch-free.

```
rtl > @ rising_edge_detector.sv > ...  
1 // Rising Edge Detector  
2 // The output signal `rise` is asserted immediately when input signal `sig_in`  
3 // transitions from low to high, and deasserted when `sig_in` becomes low OR  
4 // when the next rising edge of the clock captures the `sig_in` high signal,  
5 // whichever comes first.  
6 //  
7 // Parameters:  
8 //   None  
9 //  
10 // Ports:  
11 //   clk:      - Clock Input  
12 //   sig_in:   - Input Signal  
13 //   rise:     - Output Signal, a singular pulse within one clock cycle  
14 //  
15 // Local Variables:  
16 //   prev:     - The State that captures the previous value of `sig_in`  
17 //   next_prev: - Transitional state that captures the current  
18 //               value of `sig_in`  
19 //  
20 `timescale 1ns / 1ps  
21  
22 module rising_edge_detector (  
23     input logic clk,  
24     input logic sig_in,  
25     output logic rise  
26 );  
27  
28     logic prev = 1'b0; // State  
29     logic next_prev;  
30  
31     // Flip-Flop  
32     always_ff @(posedge clk) begin  
33         prev <= next_prev;  
34     end  
35  
36     // Next-State Logic  
37     assign next_prev = sig_in;  
38  
39     // Output Logic  
40     assign rise = (!prev && sig_in);  
41  
42 endmodule  
43
```

```

• (venv) root@4034a70d84cc:/workspace# verilator -Wall --lint-only -y rtl rtl/rising_edge_detector.sv
• (venv) root@4034a70d84cc:/workspace# iverilog -Wall -g2012 -Y .sv -y rtl rtl/rising_edge_detector.sv
• (venv) root@4034a70d84cc:/workspace# iverilog -Wall -g2012 -Y .sv -y rtl tb/wave_rising_edge_detector.v
• (venv) root@4034a70d84cc:/workspace# ./a.out
VCD info: dumpfile wave_rising_edge_detector.vcd opened for output.
tb/wave_rising_edge_detector.v:48: $finish called at 158000 (1ps)
❖ (venv) root@4034a70d84cc:/workspace#

```



```

• (venv) root@4034a70d84cc:/workspace# pytest tests/test_rising_edge_detector.py
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.0.3, pluggy-1.6.0
rootdir: /workspace
configfile: .pytest.toml
plugins: metadata-3.1.1, html-4.2.0
collected 1 item

tests/test_rising_edge_detector.py . [100%]

===== 1 passed in 0.35s =====
❖ (venv) root@4034a70d84cc:/workspace#

```

### 7.3. Button Hold Detect

```
1 module button_hold_detect #(
2     parameter int HOLD_CYCLES = 50_000_000
3 ) (
4     input  logic clk,
5     input  logic button,
6     output logic held
7 );
```

The output `held` is asserted on the clock edge at which `button` has been sampled high for `HOLD_CYCLES` consecutive rising edges, and deasserts on the clock edge after `button` goes low.

**Rationale** Entering edit mode requires the user to hold a button for a minimum duration; this module detects when that threshold has been reached.

**Implementation Notes** Instantiate a `mod_n_counter` and think carefully about its maximum count. The counter value is the entire FSM state; no additional state variables are needed.

It is more natural to reason about this counter in terms of its maximum value than its modulus  $N$ . Define `CountMax` explicitly to make the intent clear and avoid off-by-one errors.

```
1 localparam int CountMax = ... ; // Fill this in
2 localparam int CountWidth = $clog2(CountMax + 1);
3
4 logic count_rst;
5 logic count_enable;
6 logic [CountWidth-1:0] count;
7 mod_n_counter #(
8     .N(CountMax + 1),
9     .WIDTH(CountWidth)
10 ) u_counter (
11     .clk(clk),
12     .rst(count_rst),
13     .enable(count_enable),
14     .count(count)
15 );
```

From here, derive the next-state and output logic. You may find it convenient for the next-state logic to depend on the output `held`; this is permissible.

#### Report 9

Explain why the next-state logic may depend on the output `held` without violating the Moore FSM constraint.

Use `CountWidth'(CountMax)` where needed to ensure correct bit widths.

**Note** As in Section 6.6, the hold duration is exactly `HOLD_CYCLES` clock cycles only if the button is synchronised to the clock.

